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Microstructure-deformation relationship of additive friction stir-deposition Al–Mg–Si

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ABSTRACT

This research provides the first description of the process parameter relationship to the microstructure/nanostructure and mechanical properties of Aluminum Alloy 6061 AFS-D deposits. The solid-state, additive friction stir-deposition, process provides a new path for coating, joining, and additively manufacturing materials with high deposition rates (20–30 lb/hour) while avoiding liquid-solid phase transformation defects. As-deposited samples revealed the AFS-D process reduced the average grain size by an order of magnitude over a range of processing conditions. Transmission Electron Microscopy (TEM) and Atom Probe Tomography (APT) analysis established β'' precipitates were dissolved by the AFS-D process and re-precipitated as Mg–Si solute clusters.

1. Introduction

As metal manufacturing costs have increased, viable methods for producing complex geometries with high strength-to-weight ratios that produce limited waste has yielded numerous methods for metal additive manufacturing (AM) [1]. Fusion-based metal AM processes require the use of a focused, high energy source directed on to a wire [2–4] or a mixture of metallic powders [5–7] that either melts or sinters the feedstock material into consecutive layers. However, these fusion based AM processes produce intense thermal gradients that present significant challenges for traditionally difficult or unweldable alloys, namely aluminum or magnesium alloys. These fusion processes create technical barriers in producing as-deposited defect-free aluminum components with commercially available materials. In aluminum alloy 6061 (AA6061), selective vaporization can occur, resulting in change of composition from the loss of zinc, magnesium, and silicon, substantial alloying elements that characterize AA6061 precipitate strengthening and corrosion resistance [8,9]. Additionally, rapid formation of oxides [10,11] and ambient water vapor [12] requires inert atmospheres to prevent contamination and porosity during solidification, remedied by secondary solidification operations. An alternative solution to circumvent these obstacles has been to pre-alloy powders containing increased traces of silicon, scandium, or zirconium [11–13].

Microstructural analysis of fusion-based additively manufactured aluminum alloys has often produced preferential grain orientation and destruction of inherent precipitates induced by phase transformations [12,14]. Thijs et al. [15] and Croteau et al. [16] observed fine, equiaxed grains at layer interfaces giving way to columnar, preferential grains parallel to solidification gradients in the center of the melt pools. Furthermore, strong fibrous textures in the <100> direction were reported leading to anisotropic behavior of as-deposited alloys [3,13,15]. Supplemental examination on the nanoscale determined traditional precipitates were dissolved due to the vaporization of magnesium [17]. Instead, pre-alloyed elements produced cubic or triangular particles at grain boundaries on the layer interface, which pinned the migration of grains and explained the refined particle distribution seen at the top of each layer.

As-deposited aluminum alloys exhibited varying microhardness measurements depending on alloying composition and densification. Uddin et al. [13] reported Vickers hardness measurements of the most dense specimens averaging at 90 ± 6 Hv of laser powder bed fusion AA6061, but 119 ± 6 Hv with secondary tempering. When Si, Sc, or Zr was included as an alloying element, hardness varied from 93 Hv [18] to 127 ± 3 Hv [15] in the as-deposited state. The noted increase in hardness was determined to be the result of increased silicon and scandium, which produced diamond-shaped, silicon-rich precipitates and a larger distribution of Al_3Sc precipitates [7]. However, it is noted that despite the ever-improving mechanical and microscopic properties of these additive manufacturing processes, these additive powders remain difficult to produce and have been shown to require additional alloying

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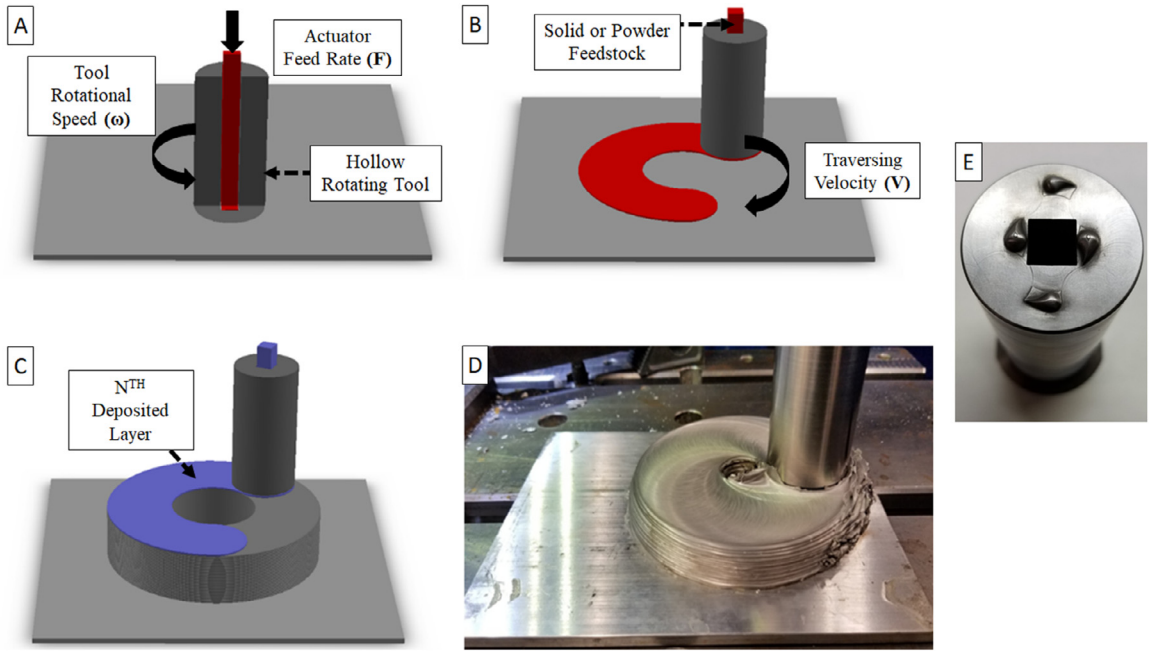

Fig. 1. (A) Cross section schematic of the beginning of the AFS-D process where the tool plunges into a substrate while spinning at a predetermined rotational speed. (B) The tool is then raised to the layer height during which time an actuator drives a solid, or powder, feedstock through the tool as the tool traverses across the substrate, producing a layer. (C) At the completion of a layer, the spindle raises to the next layer height and repeats the process to build consecutive layers. (D) Photograph of the actual AFS-D process applied to AA6061. (E) Image of the teardrop featured tool.

elements to be structurally viable. This creates a technological gap in additively manufactured, commercially available aluminum alloys and the end uses.

Additive Friction Stir-Deposition (AFS-D), commercially known as MELD, is a solid-state, AM process that benefits from similar fundamental physics occurring in friction stir welding/processing (FSW/P) [19–21]. The solid-state nature of the AFS-D process avoids deleterious effects that occur in fusion-based AM processes such as hot cracking, porosity, and columnar grain structures without the need for secondary densification post processing. As illustrated schematically in Fig. 1(A–C), robust, metallurgical bonding is achieved between deposited layers by forcing a metallic powder or solid feedstock through a rotating, non-consumable cylindrical tool. The patented tool, featuring 4 teardrop projections on the interaction face evident in Fig. 1(E), generates frictional heat and pressure inducing high-shear, severe plastic deformation of the solid feedstock onto a substrate and subsequent layers. Additionally, the severe plastic deformation generated by the tool and material interaction instigates dynamic recrystallization (DRX) in the as-deposited material, which yields a refined grain microstructure [19–21]. At the end of a predetermined path, the spindle raises by a user-defined layer height and traverses along the build route to deposit consecutive layers (Fig. 1(C)), producing a near-net shaped component (Fig. 1(D)).

To elucidate the fundamentals of the AFS-D AM process, an investigation on the influence of varying processing parameters on the deposited material microstructure is necessary. Prominent processing parameters controlling quality of parts produced include spindle rotational speed (ω), deposition feed rate (F), and traversing velocity (V) of the tool. Significant work has been conducted by the FSW/P community establishing similar effects [22–26]. By understanding the relationship between process parameters and deformation mechanisms for AFS-D, correlations can be made to effectively and efficiently deposit alternative materials at varying rates without diminishing part quality.

This is the first study to quantify the integral process-microstructure-property relationships of as-deposited AA6061 using the AFS-D process. Microhardness profiles were taken from the cross-section of builds to produce relationships between mechanical characteristics and machine parameters. Resulting average hardness values were plotted against the processing window and used to determine comparative samples for microstructural analysis. Electron backscatter diffraction (EBSD) and transmission electron microscopy (TEM) was conducted to characterize the microstructural evolution of depositions. This study provides a succinct, multiscale characterization of as-deposited AFS-D AA6061 to expound the effect of the high-shear solid-state AM process.

2. Materials and methods

A B8 MELD machine deposited square, nominally 9.53 mm solid AA6061-T651 (nominally Al, 1.16 wt% Mg, 0.51 wt% Si, 0.27 wt% Cu) feedstock rods fed through a hollow, 38.1 mm diameter, rotating tool with four “teardrop” protrusions on the tool face onto a 6.35 mm thick AA6061-T651 fixed substrate. Four 88.9 mm long consecutive layers were deposited in alternating directions at 1 mm layer heights to construct each build. As illustrated schematically in Fig. 1, tool rotational speed (ω) varying between 300 and 420 RPM, feed rates (F) ranging from 66 to 122 mm/min, and traversing velocity (V) alternating from 127 to 229 mm/min were applied as a 3-factor approach to produce each build. Fabricated cross-sections from each build were observed for lack of cohesion, voids, and galling, which assessed the success of the build for ensuing evaluation.

For microstructural analysis, successful depositions and feedstock material were ground and polished step-wise down to a 3.0 μm diamond suspension for Vickers hardness measurements. An 18 × 40 microindentation grid on a Struers Durascan at 0.2 kg indented the deposition cross-sections ensuring examination of both the as-deposited AA6061 and transient region between the deposit and substrate per ASTM E384-17 [27]. Metallographic evaluation samples were extracted from the center of each deposit and subject to an additional 1 μm diamond suspension polish followed by vibratory polish with 0.5 μm colloidal silica. A Tescan Lyra FIB-FESEM affixed with an EDAX Hikari Super EBSD camera was employed to perform EBSD grain size analysis, conducted by producing scans complying with ASTM 2627-13 [28]. Subsequently, a FEI Tecnai F-20 TEM imaged nanostructure features and the evolution

of precipitates from the feedstock and as-deposited AA6061. Deposited samples for atom probe field evaporation were fabricated by a focused ion beam (FIB) lift-out approach using a FEI Quanta Dual Beam FIB and a Lyra Tescan FIB. The tips were field evaporated in a Cameca Local Electrode Atom Probe (LEAP) 5000XS. The field evaporation process was conducted at 30 K (voltage mode). Voltage pulse fraction and frequency was 20% and 200 KHz respectively. The clusters were depicted by 1.3 at.% Mg and 1.2 at.% Si iso-concentration surface.

3. Results and discussion

Fig. 2 shows the correlation between varying process parameters and the average Vickers hardness contour plots obtained from the respective cross-sections. Specifically, in Fig. 2(A), higher average hardness values are observed as rotation speed and traversing velocity increase. Fig. 2(B) illustrates the processing boundaries between the actuator deposition feed rate (F) and traversing velocities (V). Builds that were manufactured with low material feed rates and high traversing velocities generated significant galling and voids as a result of insufficient material flow. Surface defects observed are associated with cold processing analogous with friction stir welding where insufficient material flow prevents consolidation and lack of fill on the advancing region of the tool path [29]. Conversely, operating with high material feed rates and lower traversing velocities yields reduced deposition efficiency as characterized by substantial material pushed externally from under the tool face as flash. Fig. 2(C) displays the Vickers hardness contour plot of the deposit produced by the lowest operating parameters ($\omega = 300$ RPM, $V = 127$ mm/min, 2.0 lb/hour) and Fig. 2(D) illustrates the contour plot of the deposit operating at the highest parameters in the study ($\omega = 300$ RPM, $V = 228.6$ mm/min). Fig. 2(C) and (D) represent specific samples further examined to understand the disparity seen in Fig. 2(A). The highest hardness exhibited in the build composed of the lowest processing parameters is observed in the final layers of the deposition measured at 73.6 ± 3.2 Hv which is similar to hardness results observed in FSW stir zones [30–32]. The protrusions on the tool induce material flow beyond the freshly deposited layers, resulting in

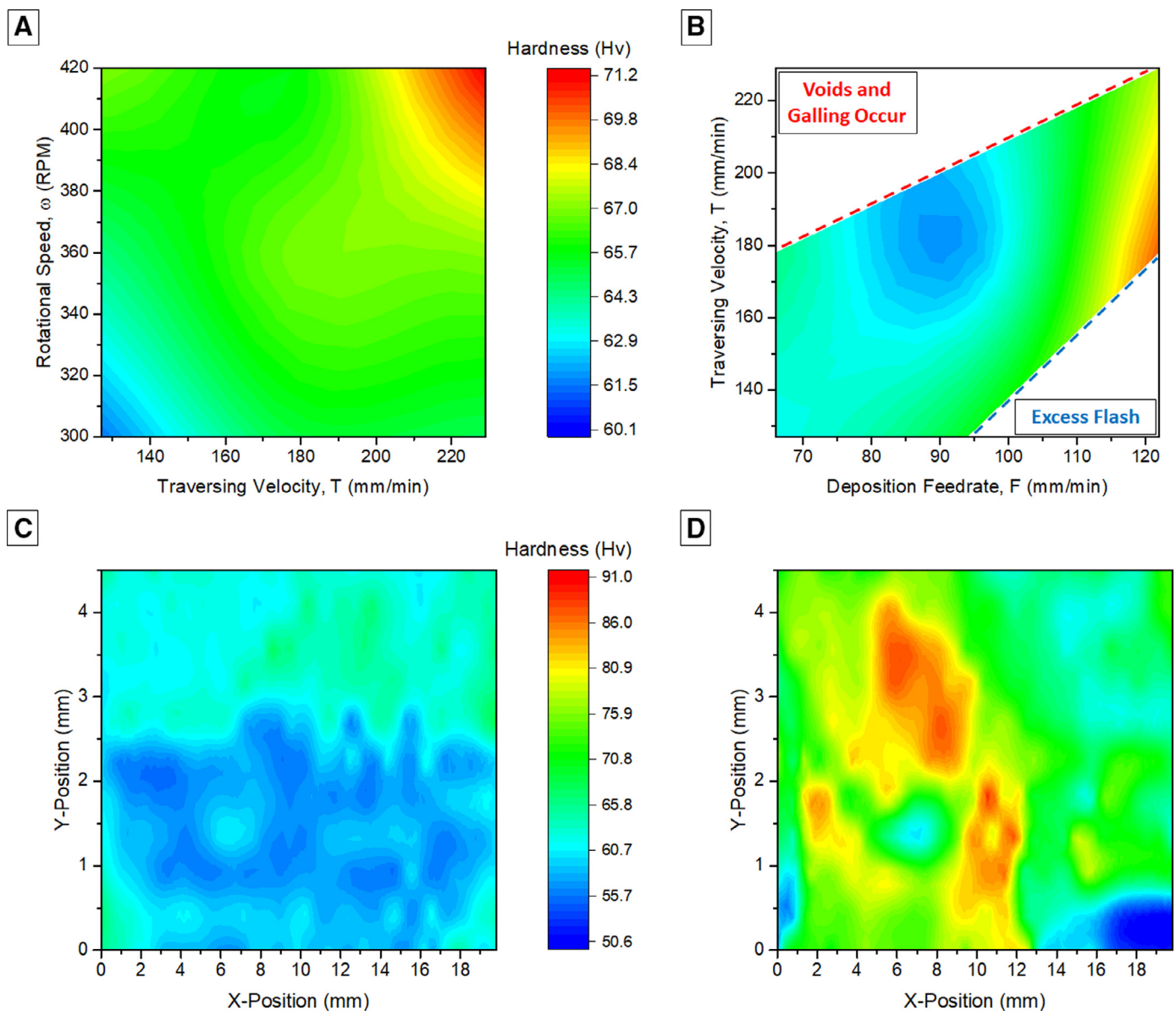

Fig. 2. (A) Parameter contour plot relating tool rotational and traversing speed to average deposition hardness. (B) Parameter contour plot relating tool traversing speed to deposition feed rate to average deposition hardness. (C) Vickers hardness plot of low-parameter AFS-D deposit. (D) Vickers hardness plot of rapid-parameter AFS-D deposit.

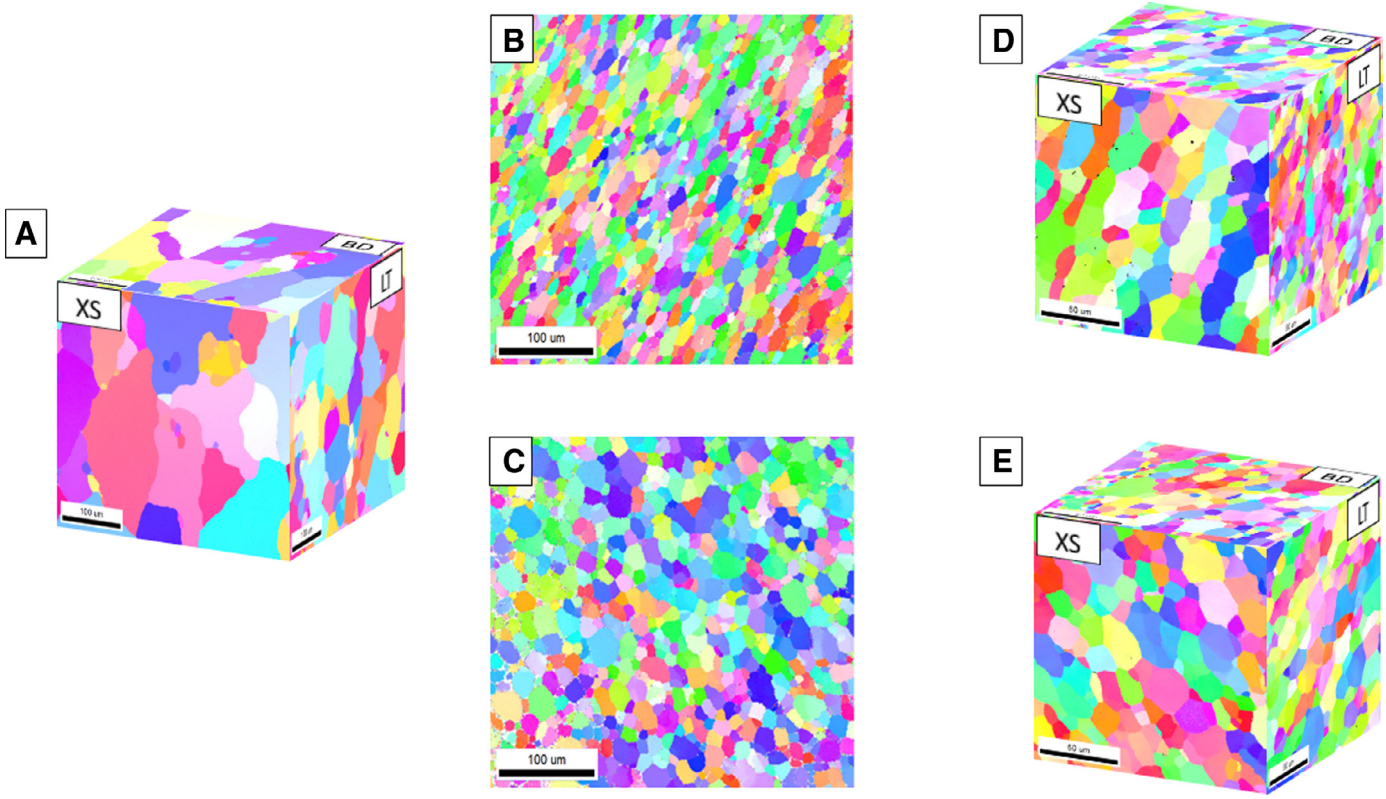

Fig. 3. (A) 3-D representative EBSD map of virgin AA6061-T651 feedstock (100 μm scale bar). (B) EBSD map of the cross section of the deposition with the slowest parameters. (C) EBSD map of the cross section of the deposition with the fastest parameters. (D) 3-D EBSD map of the slowest deposition depicting refined, equiaxed grains (50 μm scale bars). (E) 3-D EBSD map of the fastest deposition depicting refined, equiaxed grains (50 μm scale bars).

less definitive interfaces between the consecutive layers. This data is counter to results from Rivera et al. [19] due to the change from the featureless tool utilized in that study. Maximum hardness in the build with the highest investigated parameters was found to be 91.0 ± 7.0 Hv; however, the deviation of hardness values in the cross section is significantly higher than that of the other examined deposit. The less consistent nature of the hardness data in this deposit infers additional process optimization would likely improve mechanical coherency. Nevertheless, maximum hardness found in the cross sections is 20% lower than that of the feedstock at 112 ± 2.65 Hv suggesting dynamic recovery (DRV) or precipitate evolution is occurring from the intrinsic high shear mechanisms involved in the process [33], and is discussed in the proceeding sections.

Fig. 3 depicts the comparative inverse pole figure grain orientation maps for the feedstock and as-deposited AA6061. Fig. 3(A) shows the grain orientation map of the feedstock while Fig. 3(B) and (C) contrasts the angle EBSD maps and grain size transformation of the low and rapid-parameter as-deposited AFS-D specimens, respectively. ASTM grain size analysis revealed a 93% average reduction in grain size from 200 μm observed in the feedstock to 15 ± 4 μm in both as-built deposits. The substantial reduction is attributed to the DRX generated by the high normal and rotational shear strains induced by the AFS-D tool during the deformation and deposition process. Rivera et al. [19] documented a similar rate of 90% reduction in grain size in AFS-Deposited AA2219; however, the final grain size was smaller. This recrystallization mechanism is understood to be correlated to the attributes of the initial grain structure [34]. Elongated, preferential grains comprising the as-received, extruded feedstock rods were transformed into refined, equiaxed grains throughout the deposited microstructure. These grain sizes are comparable to grain sizes exhibited in AA6061 FSW weld nuggets [26,31]. Studies on traditional FSW found that minuscule decreases in both tool rotational speeds (50 RPM difference) and translational speeds (50 mm/min difference) resulted in grain sizes exhibiting a 50% greater reduction in the weld nuggets [35]. However, no such variance is observed in the evaluated AFS-deposits despite the variation in rotational and advancing speeds of 120 RPM and 102 mm/min respectively. While significant, the observed DRX is not as severe as that occurring in AFS-D of IN625, which reported ultra-fine and nanocrystalline grains [20,36]. This difference can be credited to the higher stacking fault energies inherent in aluminum alloys [19,37], and is akin to other SPD processes such as FSW and high pressure torsion [30,38]. These refined grains are equiaxed in shape as shown in Fig. 3(D, E), which present 3-dimensional inverse pole figure orientation maps collected from three orthogonal surfaces of the as-deposited AFS-D AA6061 produced with the lowest and highest defined parameter sets respectively. Thus, the process provides the driving mechanism to promote dislocations and consequently cultivating equiaxed grains.

Fig. 4 provides further supplementary visualization of the recrystallization occurring in representative grain boundary and grain reference orientation deviation (GROD) maps of the low (A, B) and rapid (C, D) parameter deposits. Low angle grain boundaries (LAGB), depicted as red misorientations from 2° to 10°, appear more frequently in the low-speed deposit and suggests a reduction in dislocation densities within grains. GROD maps further indicate this by measurement of intergranular misorientation where recrystallized grains are characterized by lower intergranular misorientation that are displayed as blue grains. By comparison, the low-parameter deposit exhibits 7.4% more recrystallized grains than that of the rapid-parameter deposit. Suhuddin et al. [39] hypothesized that misorientation distributions in FSW aluminum near the pin location were more continuous in nature by their increase in HAGB. However, it was also suggested that areas further away from the pin exhibited more discontinuous recrystallization due to the decrease in strain rate, temperature, and viscosity caused by the increased distance from the pin [40,41]. This hypothetical effect is exaggerated in AFS-D

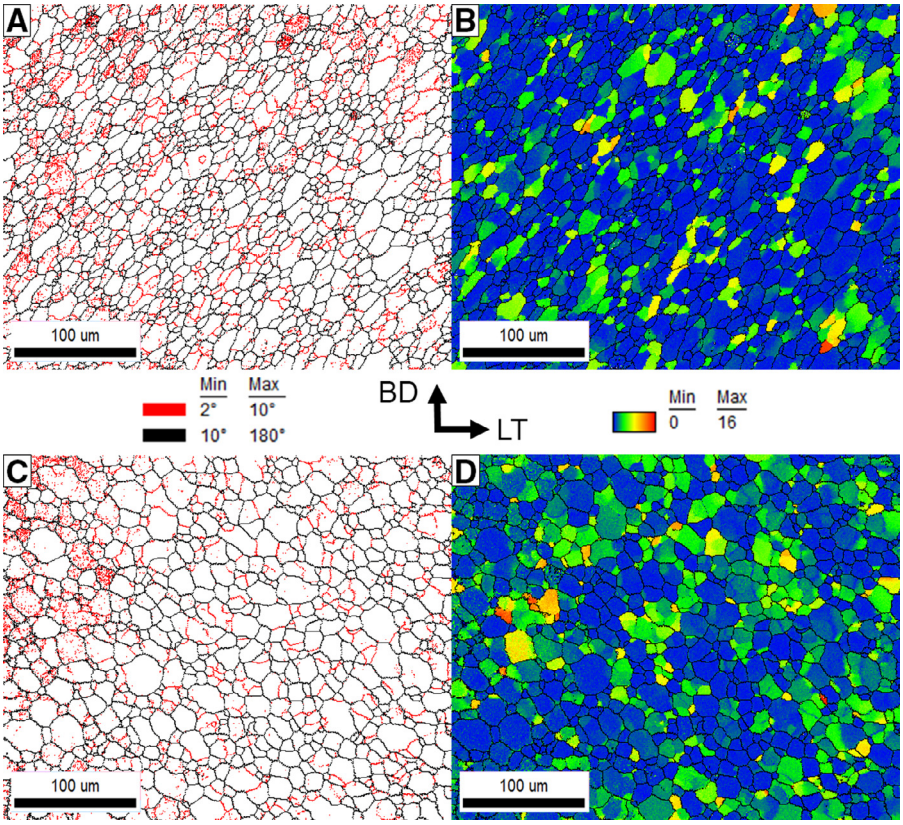

Fig. 4. Grain boundary and grain reference orientation (GROD) plots observing cross sections of the low (A, B) and rapid parameter (C, D) deposits. Low angle grain boundaries (LAGB) with misorientation angles from 2° to 10° are represented by red lines. High angle grain boundaries (HAGB) were defined as boundaries with misorientation angles greater than 10°.

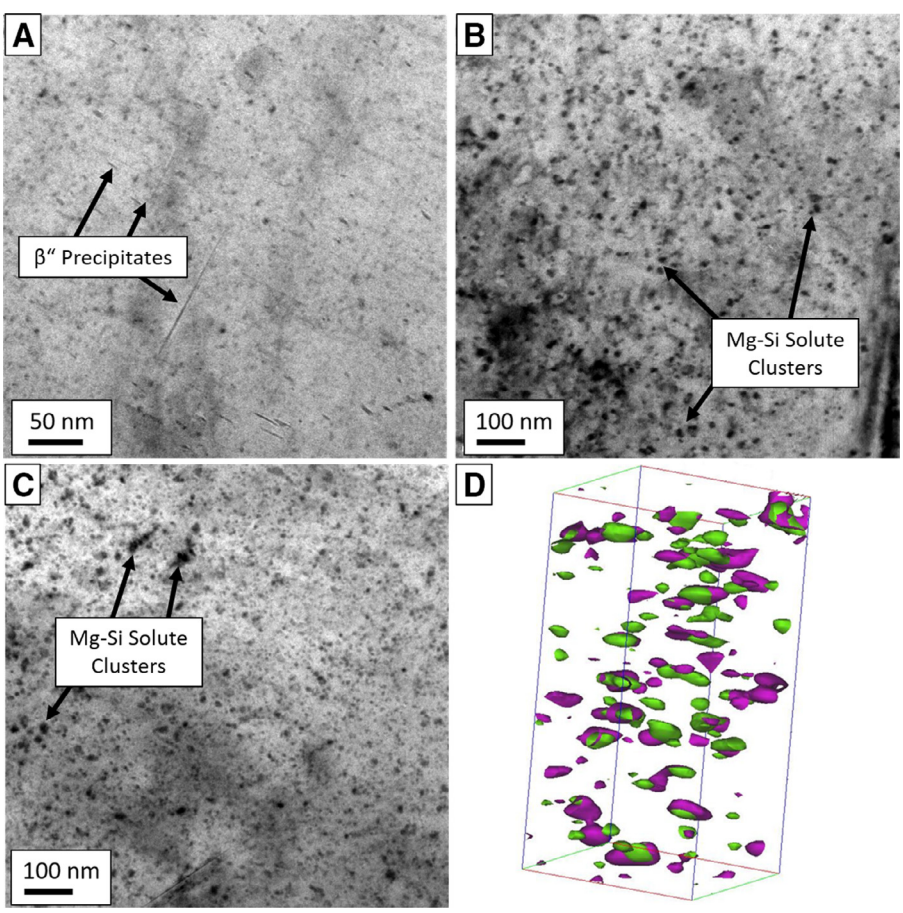

Fig. 5. (A) TEM bright field images of AA6061-T651 with conventional needle-shaped β'' precipitates. (B) TEM micrograph of the low-parameter deposit showing Mg–Si solute clusters. (C) TEM micrograph of the rapid-parameter deposit exhibiting solute clusters. (D) APT analysis visualizing Mg-Si rich clusters ($20 \times 20 \times 40$ nm^3 volume).

resulting from the lack of baseplate material ahead of the advancing tool that acts as a barrier for material flow in FSW. Consequently, that effect creates a larger wave front of deposited material, the combination of a greater deposition feed rate and lack of material pinning along the leading edge results in more discontinuous recrystallization and flashing, particularly in the rapid parameter deposit. The resulting, less consistent extent of dislocation pile-up inhibits grain movement; thus, revealing a localized higher hardness than the contrasted deposit.

Comparative microstructural examination between the feedstock material and as-deposited AFS-D precipitate behavior is depicted in bright field images in Fig. 5. Fig. 5(A) shows needle-shaped β'' precipitates characteristically found in solution heat treated and artificially aged Al–Mg–Si alloys [42] in the as-received feedstock material. Fig. 5(B,C) evaluate the precipitation behavior of the as-built AFS-D deposits, which do not exhibit the same precipitation evident in the feedstock. Rather, observations of re-precipitation occurring in the form of Mg–Si solute clusters is evidenced in the two deposits, suggesting the process dissolves β precipitates back into the aluminum matrix during the AFS-D process [19,43,44]. Atom probe tomography analysis in Fig. 5(D) provided clear details of the size and distribution of the Mg–Si rich clusters taken from the cross section of the as-deposited samples. These clusters exhibit small size (~5 nm) and circular morphology, concurrent with TEM results. This nanostructural change can be attributed to the significant shear deformation and temperatures intrinsic of the process as AA6061 traditionally undergoes precipitation transformation at 803 K (530 °C), and re-precipitation is possible as Mg–Si solute clusters at 443 K (170 °C) [44–46], similarly observed in FSW nuggets [26,31]. Additional work is necessary to examine the temperature variation while depositing various materials to effectively quantify this observation, but is estimated to be analogous with FSW where $T < T_{melt}$ [35,47]. Nevertheless, the dissolution of these traditional strengthening precipitates clarifies the reduced strength indicative in the hardness profiles, despite Hall–Petch strengthening resulting from the DRX. It is notable that despite the variance in processing parameters, the resulting microstructure in both deposits remain similar in microstructural appearance at both the micro- (grain size and morphology) and nanoscales (solute clustering).

4. Conclusions

This study presents the first investigation to elucidate the fundamental mechanisms of the solid-state additive manufacturing process, AFS-D, by quantifying the effects of varying process parameters on microstructural evolution of as-deposited AA6061 builds. The multiple deposits produced by AFS-D and cross-sectioned revealed a 20% reduction in Vickers hardness from the feedstock AA6061-T651 material. EBSD analysis revealed significant grain size reduction from 200 μm in the feedstock to an average of 15 μm in both as-deposited AA6061 deposits resulting from the inherent high shear mechanisms in the process. The microstructure additionally revealed that the slower deposit contained a more recrystallized microstructure than that of the overfed build, which may be a result of the reduced strain and viscosity occurring near the edge of the rotating tool. Subsequent TEM and APT evaluation found the dissolution of β'' strengthening precipitates and then re-precipitation of Mg–Si solute clusters in the as-deposited material, which explains the depreciated mechanical properties in the as-deposited AA6061 when compared to the feedstock. However, similarities between deposits suggests that microstructural evolution does not appear to be process dependent; thus, suggesting that visual, non-destructive inspection may be sufficient to determine deposition quality. Supplementary research is on-going to further investigate the thermal gradients involved in the process as well as a more inclusive study into mechanical characteristics of both as-deposited and post-deposited heat treated materials to promote the development of the AFS-D process.

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Declaration of interest

None.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.mtla.2019.100387.

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